

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Early computational approaches for depression detection using EEG signals primarily relied on conventional machine learning algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forests, and Linear Discriminant Analysis (LDA). In these methods, EEG signals were first pre-processed and then converted into handcrafted features such as power spectrum values, wavelet coefficients, entropy measures, and frequency band energy distributions. These extracted features were used to train classifiers for distinguishing between depressed and non-depressed subjects. Although these models achieved moderate classification performance, their effectiveness depended heavily on the quality of manually designed features, which required expertise in signal processing. Furthermore, manually extracted features often failed to capture deeper and more complex neural patterns associated with depressive states, resulting in reduced robustness and generalizability across different datasets and recording conditions.

With the advancement of deep learning, more recent models have shifted toward automated feature extraction. Convolutional Neural Networks (CNN) have been used to identify spatial and temporal characteristics in EEG signals by learning hierarchical representations directly from input data. CNN-based models eliminated the need for manual feature engineering and demonstrated improved accuracy; however, CNNs alone are limited when handling the sequential nature of EEG signals. To address this limitation, Recurrent Neural Networks (RNN) and their improved variant, Long Short-Term Memory (LSTM) networks, were introduced[11]. These models capture time-dependent changes and sequence patterns within EEG data, allowing better modeling of neural dynamics. Yet, LSTM models by themselves are not always effective at capturing localized spatial relationships across EEG channels[3].

To overcome the shortcomings of individual models, hybrid architectures combining CNN and LSTM have been introduced. CNN layers extract localized spatial-temporal patterns from EEG segments, while LSTM layers analyze signal variations over time,

enabling the model to learn both spatial and sequential dependencies[5]. These hybrid CNN-LSTM models have demonstrated superior classification performance, stability, and feature learning efficiency compared to traditional machine learning and single-network deep learning approaches.

3.2 PROPOSED SYSTEM

DepHNN (Depression Hybrid Neural Network), is designed to classify EEG signals into depressed and non-depressed states by integrating both spatial and temporal information from raw brainwave recordings. Depression is associated with altered neural synchronization and irregular electrophysiological activity in brain regions responsible for emotional regulation and cognitive processing. EEG signals therefore contain diagnostic patterns when analyzed both across channels (spatial domain) and over time (temporal domain). Since these variations are subtle, nonlinear, and subject-dependent, traditional feature engineering methods are insufficient, making a deep learning approach more appropriate for automated extraction of meaningful neural patterns.

DepHNN operates directly on raw EEG input of shape (19 channels $\times$ 256 time samples), enabling an end-to-end learning framework without dependency on handcrafted statistical or frequency-domain features. The model incorporates Convolutional Neural Networks (CNN) to learn spatial dependencies across electrodes and Long Short-Term Memory (LSTM) networks to capture temporal signal trends and rhythmic variations[14]. To prevent overfitting and ensure generalization across subjects, Dropout regularization is utilized within the dense and recurrent stages of training.

SPATIAL FEATURE EXTRACTION USING CONVOLUTIONAL LAYER

EEG signals recorded from multiple electrodes represent activities across different brain regions. Spatial differences between these channels reflect functional connectivity and neural pattern distribution. A 1D Convolutional Layer with 64 filters and a kernel size of 5 is applied across channels to learn local spatial patterns and inter-electrode relationships. The ReLU activation enhances non-linear feature mapping and stabilizes gradient behavior. This step transforms raw EEG into a structured spatial feature representation suitable for sequential modeling.

TEMPORAL DYNAMICS MODELING USING TWO-STAGE LSTM NETWORK

After spatial feature extraction, the feature matrix is reorganized through a Permute layer, aligning the data so that time steps become sequential inputs. EEG signals exhibit continuous temporal fluctuations, where characteristic rhythms and signal trajectories evolve over time. To capture these dependencies, a two-layer LSTM module is employed.

The first LSTM layer uses 64 units and returns the full sequence of hidden states, capturing long-duration temporal dependencies across the entire EEG window. The second LSTM layer, containing 32 units, compresses this sequence into a compact, high-level temporal representation. This hierarchical encoding allows the network to preserve both short-range signal fluctuations and global temporal patterns, enabling reliable recognition of depression-associated neural signatures

REGULARIZATION AND DENSE FEATURE INTEGRATION

To avoid overfitting and enhance model robustness, Dropout layers with a rate of 0.5 are applied after the LSTM block and the first dense layer. Dropout randomly disables neurons during training, preventing overdependence on specific activations and improving generalization.

The output is then forwarded to two Dense layers, each with 16 neurons and ReLU activation, to refine and transform the learned spatial-temporal representation into a linearly separable feature space. A Softmax output layer finally generates the probability distribution corresponding to the two diagnostic classes: Depressed and Non-Depressed.

CHAPTER 4

SYSTEM REQUIREMENTS SPECIFICATION